

2 Goal and Area of Application

Cell Painting (CP) is a high-content imaging and analysis technique used to capture and quantify phenotypic changes in cells induced by various perturbations, such as small molecules, as well as other modalities such as PROTACS, biologics, and genetic perturbation such as overexpression or depletion of genes. The method involves staining cells with a combination of fluorescent dyes that stain different cellular components, including the nucleus, cytoskeleton, mitochondria, endoplasmic reticulum, nucleoli, cytoplasmic RNA and Golgi apparatus. By analysing the resulting images, profiles of each condition or treatment are generated, which are informative of the cellular responses. These profiles facilitate the identification of compound mechanisms of action, potential drug targets, and off-target effects. CP involves advanced image analysis to extract and analyze thousands of morphological features, and it requires powerful data processing of the millions to billions of data points generated.

This SOP describes the process of conducting CP experiments for MoA studies. It is often the case that in phenotypic screens, the MoA of a given hit compound is unknown, and CP offers the potential to elucidate the MoA. This protocol is broadly applicable to laboratories engaged in phenotypic screening, functional genomics, and systems biology. It is particularly valuable in cases where the molecular targets of compounds are unknown or when off-target effects, polypharmacology, or compound-induced morphological heterogeneity need to be understood. CP enables the comparison of the profiles generated from images upon compound exposure, among dozens to hundreds of compound perturbations against reference profiles, making it an attractive option for MoA studies.

Importantly, **initial MoA annotations for compounds are sometimes later proven incorrect**. CP can reveal such discrepancies by providing an unbiased cellular readout. For example, a compound initially thought to act as a selective kinase inhibitor may instead cluster phenotypically with tubulin disruptors or mitochondrial poisons—uncovering hidden activities or primary mechanisms missed by targeted assays. This makes CP especially valuable for **correcting misleading MoA assignments** and for detecting unexpected biology.

CP is especially powerful for investigating the MoA of unknown or poorly characterized small molecules. By comparing morphological profiles of test compounds to those of annotated reference compounds, researchers can:

- Infer biological pathways or targets affected.
- Identify polypharmacology or off-target effects.
- Stratify compound hits from phenotypic screens based on phenotypic similarity.
- Discover structure-activity relationships not apparent from chemical structure alone.

For robust MoA inference:

- Use well-annotated reference libraries (e.g., JUMP-CP, Drug Repurposing Hub or SPECS)
- Normalize for batch effects using tools such as pycytominer
- Incorporate machine learning for classification and prediction
- Integrate with transcriptomic or proteomic profiling for multi-modal analysis

3 Definitions and Abbreviations

SOP	Standard Operating Procedure
CP	Cell Painting
MoA	Mechanism of Action

4 Method

For MoA studies, CP can be generally used as follows:

(1) Using an annotated compound library of choice to act as reference, containing labels related to at least the target and MoA, to make in silico comparisons of the molecular fingerprint from a molecule of interest to the fingerprints of the reference library, and (2) making predictions of eg side effects or toxicity, or hypothesize on what biological pathway or part of the cell is affected, by looking at observed cellular changes.

4.1 General

Below is an indicative procedure for a CP experiment as described in the original scientific work by Bray MA *et al.* Nature Methods 2016. The exact reagents, concentrations, cell lines and equipment may vary.

4.2 Information for consumables / chemicals / equipment

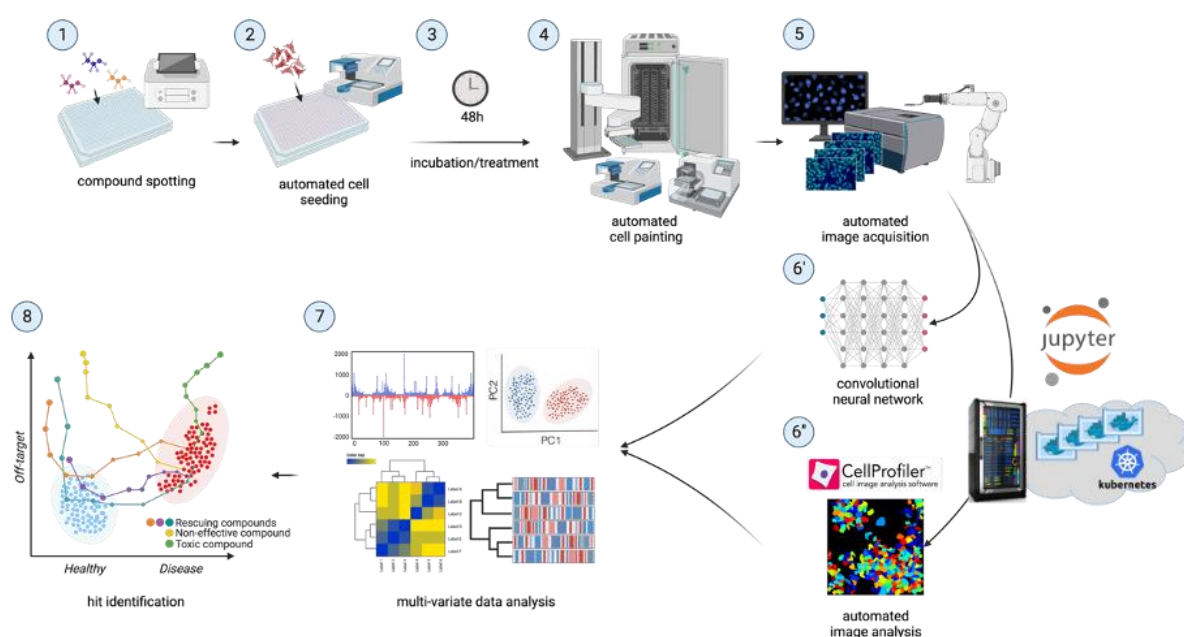
CP reagents	Target Organelle
Hoechst 33342	DNA/Nucleus
Concanavalin A	Endoplasmic Reticulum
SYTO 14	Nucleoli and cytoplasmic RNA
Phalloidin	F-actin cytoskeleton
WGA	Golgi Apparatus
MitoTracker Deep Red	Mitochondria

General reagents	
Cell lines	Typically, adherent cell lines such as: U-2 OS, A549, MRC-5 or MCF-7
Culture medium	Appropriate medium for the cell line of choice
Fixative	4% paraformaldehyde (PFA) in PBS
Permeabilization buffer:	0.1% Triton X-100 in PBS
Imaging buffer	PBS with optional Sodium Azide for long-term storage
MitoTracker incubation buffer	Hanks' Balanced Salt Solution (HBSS) or FluoroBrite

4.3 Equipment

- Multiwell plates, typically 96 or 384 (black-walled, clear-bottom)
- Automated fluorescence microscope (e.g., Molecular Devices ImageXpress, Revvity Opera Phenix, Revvity Operetta)
- Image analysis software (e.g., CellProfiler)
- Data analysis tools (e.g., R, Python, CellProfiler Analyst)
- PyCytominer for data processing and analysis
- If available, automated liquid handling will ensure better and more robust results:
 - o Robotic arm for plate handling among the instruments (e.g. Brooks PreciseFlex)
 - o Liquid dispenser (e.g. BioTek MultifloFX, ThermoFischer Multidrop, Integra WellJet)
 - o Plate washer (e.g. Tecan Hydrospeed, BioTek 405, BlueCatBio BlueWasher)
 - o Automated incubator (e.g. Liconic STX-44, ThermoFisher Cytomat 2C)

4.4 Protocol



1. Cell Seeding

- Dispense cells into the plates at an optimal density to achieve 70-80% confluency at the time of staining, this will greatly depend on the cell line and culturing medium of choice (e.g. 5000 cells/well for 96-well plates or 2000 cells/well for 384-well plates). Incubate plates at 37°C in a humidified atmosphere with 5% CO₂ typically overnight

Note: If necessary, coat plates with poly-D-lysine or another appropriate coating to enhance cell adhesion.

2. Treatment

- Prepare treatment solutions in the culture medium.
- Add compounds to wells at desired concentrations, ensuring even distribution.
- Incubation: Incubate for the appropriate duration (e.g., 24 or 48 hours, or longer).

Note: Steps 1 and 2 can be combined when using assay ready plates, in which the compounds or treatments are plated in the plates prior dispensing the cells on top.

3. Mitochondrial Staining, fixation and Permeabilization

Critical step: MitoTracker Deep Red Staining needs to be done in live cells.

- Prepare a 100 nM solution of MitoTracker Deep Red in HBSS or FluoroBrite medium.
- Add 100 μ L to each well for 96-well plates (25 μ L for 384-well plates) and incubate for 30 minutes at 37°C and 5% CO₂
- Wash: Wash twice with PBS.
- Carefully aspirate the medium without disturbing the cell monolayer.
- Add 100 μ L of 4% PFA to each well for 96-well plates (25 μ L for 384-well plates) and incubate for 15 minutes at room temperature.
- Wash: Wash twice with PBS (100 μ L for 96-well plates, 25 μ L for 384-well plates).
- Add 100 μ L of 0.1% Triton X-100 in PBS to each well for 96-well plates (25 μ L for 384-well plates) and incubate for 10 minutes at room temperature.
- Wash: Wash twice with PBS.

4. Remaining stains

Prepare a mixture with the remaining staining reagents and add 50 or 30 μ L of the mixture per well in 96 or 384 well plate, respectively. The final concentration of the stains might be optimised for each cell line and microscope, see literature references for more specific information.

5. Image acquisition

- Use an automated fluorescence microscope with appropriate filter sets for each fluorophore, typically:
 - Hoechst 33342: UV channel
 - MitoTracker Deep Red: far-red channel
 - SYTO 14: green channel
 - Concanavalin A: blue channel
 - Phalloidin: red channel
 - Wheat Germ Agglutinin: far-red channel

Note: Capture multiple fields per well (typically 5 to 9 fields) to ensure sufficient coverage and subsequent statistical power.

6. Image analysis

- Correct for background fluorescence using flat-field correction.
- Normalize illumination across the field of view.
- Tip: Use software such as Cell Profiler custom scripts for preprocessing steps.
- Utilize CellProfiler or available software for feature extraction.
- Segment cells and subcellular structures based on fluorescence intensity.
- Measure features such as area, shape, texture, and intensity for each segmented object.
- Note: Adjust segmentation parameters to optimize accuracy for different cell types and stains.
- Combine features from different stains to create a multidimensional phenotypic profile.
- Use CellProfiler Analyst or similar tools for integrated analysis.

7. Quality control

- Perform quality control to exclude wells with artifacts, insufficient cell numbers, or staining inconsistencies.
- Note: Implement automated quality control checks to streamline this process.

8. Data analysis

PyCytominer is an open source, Python-based package created by the BROAD institute that allows for the processing of Cell Painting data. This is an alternative to bespoke R or Python scripts.

- Use multivariate statistical methods such as Principal Component Analysis (PCA) or t-distributed Stochastic Neighbor Embedding (t-SNE) to analyze phenotypic profiles.
- Identify clusters of similar treatments and outliers.
- Note: R and Python offer robust libraries (e.g., scikit-learn, ggplot2) for statistical analysis and visualization.

It should be described how reagents or samples are prepared and stored during the procedure. Calculations can also be included.

9. Critical steps and aspects:

- Cell seeding density: achieve consistent and optimal confluency (~70–80%) to balance single-cell segmentation with representative phenotypic data.
- Staining quality: ensure dye stability and batch-to-batch consistency. Avoid overlapping fluorescence spectra.
- Image quality: maintain focus and minimize artifacts. Use consistent exposure settings and validate illumination correction.

- Segmentation robustness: accurate nuclear and cytoplasmic segmentation is essential. Validate segmentation outputs for each cell line.
- Plate layout: avoid edge effects and spatial biases by randomizing treatment positions.
- Batch effects: distribute replicates across imaging and processing batches and include appropriate normalization controls.

Failure in any of these steps can result in noisy or biased profiles, which reduce the assay's utility for downstream classification or MoA inference.

4.5 Additional requirements for MoA studies

It is important that to be able to compare the profiles generated from a new chemical entity to those profiles generated by using a reference library, the experiments are done in as similar conditions as possible. Of special importance is the dose of the perturbation as well as the cell line, which should be the same.

4.6 Comparison to reference set of compounds

To elucidate the potential MoA of a new chemical entity (NCE) or study a potential secondary MoA of a known and annotated drug or compound, a comparison to a reference set of annotated compounds or drugs is needed.

The reference set can range from a few dozens to thousands of compounds to which the comparison is typically performed. For example, if the biological background concerns cell death, a small reference library of a series of drugs or tool compounds inducing different types of cell death will generally suffice to provide enough data to propose a death mechanism for an NCE. On the other hand, for a broader perspective to elucidate the potential MoA of a NCE, in an agnostic manner, the comparison will benefit from a large reference library, typically thousands of annotated drugs or compounds, such as the ones curated for the Drug Repurposing Hug library at the BROAD (commercialised by SPECS).

4.7 Relating experiments for inter-project comparison

When performing MoA studies, the Cell Painting experiments should be done in the same conditions as for the experiments to generate the reference set. This includes cell line, concentration, time point and preferably protocol and microscope.

This is to ensure that the batch correction methods currently in use (and in development), can normalise the data, so that the experiments can be related to one another.

5 Further applicable resources

- Bray, MA., Singh, S., Han, H. *et al.* Cell Painting, a high-content image-based assay for morphological profiling using multiplexed fluorescent dyes. *Nat Protoc* **11**, 1757–1774 (2016). <https://doi.org/10.1038/nprot.2016.105>
- Cimini, B.A., Chandrasekaran, S.N., Kost-Alimova, M. *et al.* Optimizing the Cell Painting assay for image-based profiling. *Nat Protoc* **18**, 1981–2013 (2023). <https://doi.org/10.1038/s41596-023-00840-9>
- Francisco Rodríguez MA, Carreras-Puigvert J, Spjuth O, Designing microplate layouts using artificial intelligence, *Artificial Intelligence in the Life Sciences (2023)* Volume 3, 100073, <https://doi.org/10.1016/j.ailsci.2023.100073>.
- Ziegler S, Sievers S, Waldmann H. Morphological profiling of small molecules. *Cell Chem Biol.* 2021 Mar 18;28(3):300-319. <https://doi:10.1016/j.chembiol.2021.02.012>
- Caicedo, J., Cooper, S., Heigwer, F. *et al.* Data-analysis strategies for image-based cell profiling. *Nat Methods* **14**, 849–863 (2017). <https://doi.org/10.1038/nmeth.4397>
- Seal, S., Trapotsi, MA., Spjuth, O. *et al.* Cell Painting: a decade of discovery and innovation in cellular imaging. *Nat Methods* **22**, 254–268 (2025). <https://doi.org/10.1038/s41592-024-02528-8>